

VR cues can mislead disabled users into hazardous movements [204]. The authors in [34] systematically investigate how multimodal LLMs paired with AR devices can be weaponized for next-generation social engineering, introducing the SEAR framework. The SEAR framework pairs multimodal LLMs with AR devices for social engineering by fusing visual and audio context, retrieving the target’s digital footprint, and driving an agent through conversational stages.

6.6 Autonomous Agent Networks

Finally, attacks in autonomous agent networks include knowledge poisoning, prompt injection, backdoored system prompts, adaptive jailbreaks, and misinformation flooding. LLM agents execute attacks by crafting malicious prompts, corrupting memory, and amplifying errors through collaboration. Debar *et al.* in [50] outline threats when nodes can explain, plan, and act. Tete *et al.* in [183] provide a taxonomy for agent applications, focusing on backdoored prompts. Ju *et al.* in [100] show misinformation can flood multi-agent communities within minutes, reducing task success by 42%. Pasquini *et al.* in [146] reveal benign prompt-injection can defend against LLM hacking, while Wang *et al.* in [191] use reinforcement learning for adaptive jailbreaks. Agent-native networks are both attacker and defender domains, requiring formal verification and memory isolation. Countering these threats requires hardening reasoning integrity, controlling memory updates, and ensuring prompt sanitization.

6.7 Lessons Learned for Blue Teams

- (1) **Trust and Reputation Mechanisms:** In infrastructure-free environments, LLM-based agents can create fake identities to conduct Sybil attacks and manipulate consensus. Blue teams must implement trust mechanisms like cryptographic attestations and behavioral scoring to ensure network accountability.
- (2) **Resilience Through Redundancy and Decentralized Recovery:** LLM-based agents can target weak points in peer-to-peer networks to disrupt communication. Blue teams should design networks with redundancy in routing, storage, and decisions, and incorporate decentralized recovery protocols can help maintain function under compromise.

7 Future Research Directions

1) Governance/Guardrails for LLM-based Agents: Developing effective governance for LLM-based agents is critical. Unlike traditional tools, these agents can reason and escalate attacks independently. To mitigate risks, agent architectures must embed safety constraints. Research should implement ethical enforcement, compliance checking, and intervention mechanisms. Standardized audit frameworks would ensure transparency and accountability. International policies must regulate agents while preserving innovation.

2) Human-in-the-Loop Alignment for LLM-based Cyberattack Agents: As LLM-based agents acquire increasing autonomy, integrating human oversight becomes a fundamental challenge [140]. Systems should ensure human review at critical decision points during high-risk operations. Research must balance autonomy and human intervention while maintaining effectiveness. Dynamic human-in-the-loop systems and reinforcement learning from feedback can support this goal. Agents should seek human guidance when encountering ethical ambiguities, creating a symbiotic relationship between human expertise and machine operation.

3) Sustainable Red-teaming: Red-teaming uses simulated adversaries to test vulnerabilities while accounting for environmental impact [65]. Research should develop efficient methodologies that minimize energy use while maintaining vulnerability coverage. Techniques like scenario sampling, model distillation, and RL-based exploration can improve resource efficiency. Sustainable red-teaming practices will enhance both AI safety and environmental responsibility.

4) Privacy-preservation during Multi-Agent Collaboration: Federated learning enables collaborative improvement without centralized data collection [41]. Future research should explore protocols for agents to share threat insights while protecting organizational data. Key challenges include secure aggregation, poisoning resistance, and non-IID data robustness. Real-time federated updates could help defensive agents quickly adapt to new attack patterns.

5) Defense Against LLM-based Agent Swarms: As single-agent threats evolve into coordinated multi-agent attacks, future defenses must prepare for the possibility of intelligent agent swarms executing synchronized cyber operations [161]. Future research should focus on developing detection and mitigation techniques specifically tailored to the behavioral signatures of swarm-based attacks. Distributed anomaly detection, decentralized defense architectures, and deception-based countermeasures capable of confusing or fragmenting swarm coordination will be vital. Defensive swarms composed of autonomous security agents could also be explored as a countermeasure, creating dynamic, self-organizing barriers against distributed attacks at machine speed.

6) LLM-based Agent Honeypots: Deception remains a powerful tool in cybersecurity, and the emergence of LLM-based agents unlocks new possibilities for intelligent, adaptive honeypots [134, 139]. Future honeypots could leverage LLM capabilities to engage attackers in realistic dialogues, simulate system behaviors dynamically, and capture detailed telemetry of attack tactics. Developing efficient, scalable LLM-based honeypots could transform cyber defense from a reactive model into a proactive intelligence-gathering operation.

7) Agent-to-Agent Deception: Cyber conflict now includes autonomous adversarial agents [214]. Deception between LLM-based agents is a crucial research frontier. Defensive strategies could deploy decoys and misinformation to mislead attacker agents. Researchers must also defend against malicious agents manipulating defensive AI. Agent-to-agent cyber deception will require interdisciplinary insights from game theory, adversarial machine learning, and multi-agent systems to craft effective tactics and countermeasures.

8 Conclusion

This survey highlights a fundamental shift in the cybersecurity landscape, driven by the rise of autonomous LLM-based cyberattack agents. These agents make sophisticated cyber threats more scalable, more accessible, and more difficult to defend against. As attack costs fall and operational complexity increases, traditional defenses are struggling to keep pace. The spread of coordinated multi-agent systems further amplifies the challenge. To respond, the cybersecurity community must adopt forward-looking strategies that prioritize adaptability, intelligent defense, and proactive threat engagement. Ultimately, understanding the strategic implications of LLM-enabled threats is essential to safeguarding the future of digital infrastructure.

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